THE DIRICHLET TENSOR OF TWO POLYNOMIAL COMONOIDS IS THE PRODUCT INTERNAL CATEGORY — IN EVERY FINITELY COMPLETE CATEGORY

MACBETH

Abstract. In **Set**, comonoids for the substitution product $\triangleleft$ of polynomial functors are exactly small categories, and the Dirichlet tensor $\otimes$ of two such comonoids carries a comonoid whose represented category is the product $C \times D$ on the nose — the **Set** case is Niu–Spivak [2, Prop. 8.79]. We show this is not an artefact of **Set**, nor of any size, coproduct, or well-pointedness hypothesis. Let $\mathcal{E}$ be any finitely complete category and C, D internal categories in $\mathcal{E}$ whose domain maps are exponentiable, so that they are $\triangleleft$-comonoids c, d in the category $\mathrm{Poly}(\mathcal{E})$ of internal polynomials in the sense of Shapiro–Spivak [3]. Then $c \otimes d$ carries a canonical $\triangleleft$-comonoid structure whose represented internal category is the product $C \times D$ in $\mathbf{Cat}(\mathcal{E})$ — componentwise objects, morphisms, identities and composition — *on the nose*. Beyond the ambient Shapiro–Spivak polynomial framework the proof uses only finite limits: its single load-bearing coherence is the commutation of products with pullbacks, a finite-limit fact that needs no local cartesian closure, no coproducts, no extensivity and no well-pointedness. The statement therefore holds uniformly across every elementary and Grothendieck topos and every locally cartesian closed category, with **Set** as the degenerate case. We separate carefully this (true) identification of the represented category from the (false) assertion that $\otimes$ is the categorical product of comonoids: among comonoids — equivalently cofunctors — $\otimes$ is a monoidal, non-cartesian product, and the genuine categorical product lives on the functor side $\mathbf{Cat}(\mathcal{E})$. A worked witness in the presheaf topos **Graph** illustrates the identification and is machine-checked in Lean.

1. Introduction

The equivalence chain, and whether it is a feature of Set. A recurring discovery of the last fifteen years is that four superficially different notions of "directed shape" coincide:

$$\text{containers} \;\simeq\; \text{directed containers} \;\simeq\; \text{polynomial comonoids} \;\simeq\; \text{small categories}.$$

The last of these equivalences is the one this note is about. Write Poly for the category of polynomial endofunctors on **Set**, equipped with the substitution (composition) monoidal product $\triangleleft$ with unit $y = \mathrm{Id}$. A theorem of Ahman and Uustalu, given its modern comonoid phrasing by Spivak and collaborators [4, 2], identifies the comonoids in $(\mathrm{Poly}, \triangleleft, y)$ with small categories:

$$\mathrm{Comon}(\mathrm{Poly}, \triangleleft, y) \;\simeq\; \mathbf{Cat}.$$

A polynomial carries a *second* monoidal product, the Dirichlet tensor $\otimes$, which in **Set** is the Day convolution induced on Poly by the cartesian structure $(1, \times)$ — Niu–Spivak [2, Prop. 3.79]. The two products interlock into a normal duoidal structure, so $\triangleleft$-comonoids are closed under $\otimes$; and a short computation shows that the resulting comonoid on $c \otimes d$ is precisely the product category $C \times D$. This last fact, in **Set**, is Niu–Spivak [2, Prop. 8.79]: carrier $c \otimes d$, counit $\varepsilon_c \otimes \varepsilon_d$, comultiplication through the duoidal interchanger, with codomains and composition coordinatewise.

Date: 23 October 2026.

[0]MacBeth (`scotmacbeth`) · 2026-10-23 · for Rick · content commit `PENDING`. Theorem 1.1 (the on-the-nose identification) is proved here, as a short corollary of the Shapiro–Spivak polynomial framework [3]; the **Set** case is Niu–Spivak [2, Prop. 8.79]. The **Graph** witness of §6 is machine-checked (https://github.com/scotmacbeth/work-in-progress/blob/main/lean/2026-10-21-day-convolution-graph.lean, Mathlib-free, `[propext]` only) and is offered as an illustration, not a result.

This is a clean fact about **Set**. It is natural to ask whether it is *only* a fact about **Set**. The question is not idle. The entire value of the equivalence chain, as a foundation for compositional systems, is that it should be a theorem about *structure*, not about a privileged ambient universe; and the concrete compositional models one wants — presheaves on a site of locations, sheaves on a space of states, graphs of a network — live in categories other than **Set**. If the Dirichlet-tensor-is-product fact ceased to hold the moment one left **Set**, the chain would be a **Set**-level coincidence rather than a transportable principle.

The main result. It is not a coincidence. The identification transports, verbatim, to *any finitely complete category* in which the ambient polynomial framework is available — which, following Shapiro–Spivak [3], asks only that the relevant maps be exponentiable.

Theorem 1.1 (On-the-nose identification)**.** *Let $\mathcal{E}$ be a finitely complete category, and let C, D be internal categories in $\mathcal{E}$ whose domain maps $\mathrm{dom}_C, \mathrm{dom}_D$ are exponentiable, with associated $\triangleleft$-comonoids c, d in $\mathrm{Poly}(\mathcal{E})$. Then $c \otimes d$ carries a canonical $(\triangleleft, y)$-comonoid structure — counit $\varepsilon_c \otimes \varepsilon_d$, comultiplication $\iota \circ (\delta_c \otimes \delta_d)$ via the normal-duoidal interchanger ι of* [3, Thm. 4.5] *— and the internal category it represents is the product internal category $C \times D$,* on the nose*: object-of-objects $C_0 \times D_0$, object-of-morphisms $C_1 \times D_1$, with domain, codomain, identities and composition all componentwise. In symbols, $c \otimes d = (C \times D)^\sharp$ in $\mathrm{Comon}(\mathrm{Poly}(\mathcal{E}), \triangleleft, y)$.*

Over **Set** this is [2, Prop. 8.79] word for word. The internal statement does not appear in [3] or [2]; it is a short corollary of their results — normal duoidality [3, Thm. 4.5] plus the comonoid–category identification [3, Thm. 5.6, Rem. 5.7] — and that is how we present it. We claim the internal corollary and the honest accounting around it; we do not claim either input.

What is, and what is not, a product. One warning belongs on the first page, because an earlier version of this note got it wrong. The identification $c \otimes d = (C \times D)^\sharp$ says that the *represented internal category* of $c \otimes d$ is $C \times D$. It does *not* say that $c \otimes d$ is the categorical product of c and d in the category of comonoids. That stronger statement is *false*, already in **Set**. Comonoid morphisms in Poly are *cofunctors*, not functors [3, Cor. 5.12, Def. 5.11]; and among cofunctors $C \times D$ is not the categorical product — as Niu–Spivak note, the usual product of categories "is not the categorical product in $\mathbf{Cat}^\sharp$; it is, however, a monoidal product on $\mathbf{Cat}^\sharp$, coinciding with the parallel product" [2, §8.2.4]. We give the two-line counterexample in §4. The genuine categorical product of C and D lives on the *functor* side: in $\mathbf{Cat}(\mathcal{E})$, where morphisms are internal functors, $C \times D$ is the product, by the elementary fact that $\mathbf{Cat}(\mathcal{E})$ has all limits $\mathcal{E}$ has. That is a statement about $\mathbf{Cat}(\mathcal{E})$, not about $\otimes$ on comonoids, and we keep the two apart.

Why "on the nose", and what the proof does not use. The phrase "on the nose" is doing real work. Morphisms of $\mathrm{Poly}(\mathcal{E})$ are iso-classes of diagrams [3, Def. 3.7, 3.13], so an equality of polynomials is a genuine equality of such classes; the claim is that the structure maps of $c \otimes d$ — read off through the comonoid-to-category dictionary — are *equal*, in this sense, to the structure maps of $C \times D$, with the only isomorphism in sight being the canonical one witnessing that products commute with pullbacks. The reason is a clean division of labour. The non-trivial, non-pointwise ingredient of the substitution product $\triangleleft$ (it uses the dependent product Π along domain maps, hence their exponentiability — it is *not* computed objectwise in a presheaf topos) is absorbed entirely into the ambient framework and into the comonoid-to-category translation, and never resurfaces in the identification. What remains, once one is speaking the language of internal-category data $(C_0, C_1, \mathrm{dom}, \mathrm{cod}, \mathrm{id}, \mathrm{comp})$, is componentwise finite limits.

Three ingredients the reader might expect to see are conspicuously absent:

- *No well-pointedness.* The counit $\varepsilon_c : c \to y$, which in the **Set** picture "picks the identity id_i in each hom-set", is here literally the single identity-assigning section $\mathrm{id} : C_0 \to C_1$ of $\mathcal{E}$ — not a choice of global elements. This is the point at which every naive **Set**-level

argument would break, and it does not break. The **Graph** witness of §6 is built with a *loopless* object-of-objects, which has no global element at all, to make the point concrete.

- *No local cartesian closure for the identification itself.* Exponentiability of the domain maps is what makes $\mathrm{Poly}(\mathcal{E})$, $\triangleleft$ and $\otimes$ exist at all [3, Thm. 3.26, Def. 4.1, Lem. 4.3, Thm. 4.5]; but the *identification* $c \otimes d = (C \times D)^{\sharp}$ consumes, beyond that ambient framework, nothing more than finite limits. Its one coherence is

$$(\star) \qquad (C_1 \times_{C_0} C_1) \times (D_1 \times_{D_0} D_1) \;\cong\; (C_1 \times D_1) \times_{C_0 \times D_0} (C_1 \times D_1),$$

  the commutation of products with pullbacks — a finite-limit fact, true in any finitely complete category, needing no dependent products, no coproducts, and no extensivity.
- *No extensivity, no disjointness of coproducts.* None is used; $(\star)$ is a limit–limit interchange, not an extensivity axiom.

So the honest reach of the method is the Shapiro–Spivak setting itself: any finitely complete category with the exponentiable maps their framework requires. Every locally cartesian closed category and every elementary topos — Grothendieck or not, with or without a natural-numbers object — is an instance, and so is **Set**.

A question that inverted into its own answer. This note is the positive residue of a negative-looking investigation. For some months the program pursued a "which base V?" question: over which monoidal bases does the product-is-Dirichlet-tensor phenomenon survive? The characteristic failure case was $V = \mathbf{Vec}$. Read under "for *all* V?", the failure looks like a dead end. Read under "for *which* V?", it is informative: the phenomenon asks for a base with genuine dependent-type structure — enough that the maps carrying the directions of a polynomial are exponentiable. A finitely complete category with that much structure supplies it, and the present theorem states the positive side: the fact does not merely survive the passage away from **Set**, it is *uniform* across every such base. We do not overclaim the location of the **Vec** obstruction: **Vec** is not locally cartesian closed and its addition maps are not the exponentiable direction-bundles the framework wants, so $\mathrm{Poly}(\mathbf{Vec})$ has no standard construction — but the Shapiro–Spivak hypotheses are weaker than local cartesian closure, so we claim no sharp dividing line, only that cartesianness of the tensor was never the issue.

Outline. §2 recalls $\mathrm{Poly}(\mathcal{E})$, the two products, the comonoid–category dictionary, and the interchanger, over a finitely complete base with exponentiable maps. §3 proves Theorem 1.1 in three steps: the underlying object (finite limits), the interchanger (cited), and the structure trace (the finite-limit coherence $(\star)$). §4 records, with a two-line counterexample, that $\otimes$ is *not* the categorical product of comonoids. §5 isolates, in a table, exactly what the base must supply. §6 gives the machine-checked **Graph** illustration. §7 states the honest bounds.

2. Preliminaries

We fix a finitely complete category $\mathcal{E}$. Following Shapiro–Spivak [3], the polynomial framework requires no more: a *polynomial* is an *exponentiable* morphism, and all of $\mathrm{Poly}(\mathcal{E})$, the substitution product $\triangleleft$, the Dirichlet tensor $\otimes$ and the normal duoidal structure are available over any such $\mathcal{E}$. Recall that a morphism $s : B \to A$ is exponentiable when the pullback functor $s^* : \mathcal{E}/A \to \mathcal{E}/B$ has a right adjoint Π_s (dependent product along s); in a locally cartesian closed category *every* morphism is exponentiable, and in an elementary topos likewise, but we never assume this globally.

2.1. Internal polynomial functors.

Definition 2.1 ([1, 3])**.** A *polynomial* p in $\mathcal{E}$ is an exponentiable morphism (a *bundle*) $s_p : B_p \to A_p$ of $\mathcal{E}$. Its *positions* object is A_p and its *directions* bundle is s_p; the *fibre* over a generalized element

I of A_p is $p[I] := s_p^{-1}(I)$. The associated endofunctor $\mathcal{E} \to \mathcal{E}$ is

$$p(X) \;=\; \Sigma_{A_p}\, \Pi_{s_p}\, B_p^* X,$$

where $B_p^* X = X \times B_p$ (pullback of $X \to 1$ along $B_p \to 1$) and Σ_{A_p}, Π_{s_p} are dependent sum and product. Over a locally cartesian closed base Gambino–Kock [1] first constructed the category $\mathrm{Poly}(\mathcal{E})$ with its *substitution* (composition) product $\triangleleft$ and unit $y := (\mathrm{id} : 1 \to 1)$; Shapiro–Spivak [3, Thm. 3.26] give the same product over any finitely complete $\mathcal{E}$ for polynomials whose bundles are exponentiable. Morphisms run *forward* on positions and *backward* on directions.

The construction of $\triangleleft$ is where exponentiability is spent: composing the functors $p \circ q$ and re-presenting the composite as a polynomial requires the dependent product Π along a direction bundle. The identification of §3 does *not* revisit this; it works at the level of internal-category data, where only finite limits occur.

Definition 2.2 (Dirichlet tensor [3, Def. 4.1])**.** The *Dirichlet tensor* of bundles $p = (B_p \to A_p)$ and $q = (B_q \to A_q)$ is the product bundle

$$s_p \times s_q \;:\; B_p \times B_q \;\longrightarrow\; A_p \times A_q,$$

formed with the finite products of $\mathcal{E}$ (exponentiability is preserved by products, [3, Lem. 4.3]). Its positions object is $A_p \times A_q$ and its fibre over (I, J) is $p[I] \times q[J]$. On representables $y^A \otimes y^B = y^{A \times B}$; its unit is again y. Over **Set** this $\otimes$ is the Day convolution on Poly induced by $(1, \times)$, [2, Prop. 3.79]; over a general $\mathcal{E}$ we take ($\star$)'s ambient definition — $\otimes$ is *the product of bundles* — and make no Day-convolution claim (see §7).

2.2. **The comonoid–category dictionary.** A $\triangleleft$-comonoid is a triple (c, ε, δ) with $c = (s_c : B_c \to A_c)$, counit $\varepsilon : c \to y$ and comultiplication $\delta : c \to c \triangleleft c$, satisfying counit and coassociativity. Unwinding over $\mathcal{E}$ (every "element of a fibre" read as a generalized element / section / $\mathcal{E}$-morphism):

- $\varepsilon : c \to y$ is, forward on positions, the unique map $A_c \to 1$, and backward on directions a *section* $e : A_c \to B_c$ of s_c. Dictionary: $A_c = C_0$ (object-of-objects), $B_c = C_1$ (object-of-morphisms), $s_c = \mathrm{dom}$, and $e = \mathrm{id} : C_0 \to C_1$ the identity-assigning section.
- $\delta : c \to c \triangleleft c$ is, forward on positions, the Π_{dom}-transpose of cod — a map $C_0 \to A_{c \triangleleft c}$ whose global content is $\mathrm{cod} : C_1 \to C_0$ — and backward, over the pullback of composable pairs $C_1 \times_{\mathrm{cod}, C_0, \mathrm{dom}} C_1$, the composition $\mathrm{comp} : C_1 \times_{C_0} C_1 \to C_1$.

The comonoid laws are exactly the internal-category axioms (left/right identity, associativity). This is the internal form of the Ahman–Uustalu equivalence; its *full* form over a finitely complete $\mathcal{E}$, both directions and a one-by-one match of structure equations on unconstrained data, is Shapiro–Spivak [3, Thm. 5.6, Rem. 5.7]. We use only the *forward* direction — internal category $\rightsquigarrow$ comonoid, a direct construction — for C, D and $C \times D$.

Remark 2.3 (No well-pointedness). The counit's backward datum is the single section $\mathrm{id} : C_0 \to C_1$ of $\mathcal{E}$. No global elements of C_0 are chosen, needed, or available in general. The **Set**-level gloss "ε picks $\mathrm{id}_i \in c[i]$" is, internally, this one morphism. §6 illustrates the point against a C_0 with no global element whatsoever.

2.3. **The normal-duoidal interchanger.** $(\mathrm{Poly}(\mathcal{E}), \otimes, \triangleleft, y)$ is a *normal* duoidal category [3, Thm. 4.5]: the units of $\otimes$ and $\triangleleft$ coincide (both y), so the unit-comparison maps are identities and the only non-trivial structure map is the lax interchanger

$$\iota :\; (p \triangleleft p') \otimes (q \triangleleft q') \;\longrightarrow\; (p \otimes q) \triangleleft (p' \otimes q'), \qquad \text{natural in } p, p', q, q', \tag{1}$$

given by [3, Thm. 4.5, eq. (16)] (the **Set** formula is [3, Ex. 4.6]). Its coherence is cited, not load-bearing: the comonoid laws of $c \otimes d$ follow in §3 from the on-the-nose identification, independently of the abstract coherence of ι. On positions, ι is the canonical comparison $\Pi_X A \times \Pi_Y B \to \Pi_{X \times Y}(A \times B)$ (fibrewise $A^X \times B^Y \to (A \times B)^{X \times Y}$); we will only use that it sends the pair of transposes of $\mathrm{cod}_C, \mathrm{cod}_D$ to the transpose of $\mathrm{cod}_C \times \mathrm{cod}_D$, by naturality of transposition.

3. The identification $c \otimes d = (C \times D)^\sharp$

We prove Theorem 1.1. Fix internal categories C, D with comonoids $c = (\mathrm{dom}_C : C_1 \to C_0)$ and $d = (\mathrm{dom}_D : D_1 \to D_0)$, and set

$$\varepsilon_{c\otimes d} := (\varepsilon_c \otimes \varepsilon_d) \qquad (\text{into } y \otimes y \cong y, \text{ using } 1 \times 1 \cong 1), \qquad \delta_{c\otimes d} := \iota \circ (\delta_c \otimes \delta_d).$$

Step 1 — the underlying object, from finite limits alone. By the definition of $\otimes$ as the product of bundles, the underlying bundle of $c \otimes d$ is the product map

$$\mathrm{dom}_C \times \mathrm{dom}_D \ : \ C_1 \times D_1 \ \longrightarrow \ C_0 \times D_0.$$

This is, on the nose, the underlying bundle of $(C \times D)^\sharp$, whose domain map is by definition $\mathrm{dom}_{C\times D} = \mathrm{dom}_C \times \mathrm{dom}_D$. The fibre identity $(B_p \times B_q)_{(I,J)} = p[I] \times q[J]$ is the universal property of a product of maps. Hence $c \otimes d$ and $(C \times D)^\sharp$ have equal underlying objects of $\mathcal{E}$.

Step 2 — the interchanger. Instantiate (1) at $p = p' = c$, $q = q' = d$. On positions, ι sends $\big((I, \varphi), (K, \psi)\big) \mapsto \big((I, K), \varphi \times \psi\big)$, a product of $\mathcal{E}$-morphisms; this is the position-level comparison $\Pi_X A \times \Pi_Y B \to \Pi_{X\times Y}(A \times B)$ recalled in §2.3. On directions (backward), ι reassociates the composable-pair objects, and the content of the reassociation is the finite-limit coherence ($\star$). The interchanger itself is cited from [3, Thm. 4.5]; the only fact we verify by hand is that ($\star$) is a genuine isomorphism, which it is in any finitely complete category because products commute with pullbacks. No local cartesian closure, no extensivity, no disjointness of coproducts, and no cartesianness of $\otimes$ is invoked here.

Step 3 — the structure trace. Tracing $\varepsilon_{c\otimes d}$ and $\delta_{c\otimes d}$ through Steps 1–2, with every "element" an $\mathcal{E}$-morphism:

- *Codomain.* $(\delta_c \otimes \delta_d)$ carries, on positions, the pair of Π-transposes of cod_C and cod_D; by naturality of transposition ι sends this to the transpose of $\mathrm{cod}_C \times \mathrm{cod}_D$. As an $\mathcal{E}$-morphism, $\mathrm{cod}_{c\otimes d} = \mathrm{cod}_C \times \mathrm{cod}_D : C_1 \times D_1 \to C_0 \times D_0$.
- *Composition.* On directions, ι reassociates via ($\star$) and $(\delta_c \otimes \delta_d)$ applies $\mathrm{comp}_C, \mathrm{comp}_D$ componentwise, giving $\mathrm{comp}_{c\otimes d} = \mathrm{comp}_C \times \mathrm{comp}_D$ under the canonical isomorphism ($\star$).
- *Identities.* $\varepsilon_c \otimes \varepsilon_d$ yields the section $\mathrm{id}_C \times \mathrm{id}_D : C_0 \times D_0 \to C_1 \times D_1$.

These are, morphism for morphism, the structure maps of the internal product category $C \times D$. Hence $(c \otimes d, \varepsilon_{c\otimes d}, \delta_{c\otimes d}) = (C \times D)^\sharp$ as comonoids. Its comonoid laws are those of the genuine internal category $C \times D$ (a category because C, D are, and because products of internal categories are internal categories — §4); and that these are matched by the comonoid laws under the dictionary is the one-by-one correspondence of structure equations established in the proof of [3, Thm. 5.6], not merely a bijection of lawful structures. This proves Theorem 1.1. □

Remark 3.1 (A three-line alternative). Shapiro–Spivak's own technique gives a second proof. By [3, Lem. 5.5] comonoid and internal-category structure both transport along the fully faithful singular functor $\mathcal{E} \hookrightarrow \widehat{\mathcal{E}}$ into presheaves, where the category side is computed pointwise; pointwise, the claim is the **Set** case [2, Prop. 8.79], applied at each object. Full faithfulness returns the result to $\mathcal{E}$. We have given the direct trace instead because it exhibits exactly which finite limit ($\star$) is at stake.

Remark 3.2 (No smuggled assumptions). The proof uses no choice (only chosen pullbacks enter, and the dependent-product transposes are canonical), no excluded middle (there is no case split or decidability appeal), and no pointwise coend (none is computed; $\otimes$ is the product of bundles). The element notation throughout is legitimate as generalized elements.

4. The tensor is not the categorical product of comonoids

Theorem 1.1 identifies the *represented internal category* of $c \otimes d$ with $C \times D$. It is tempting to upgrade this to "$c \otimes d$ is the categorical product of c and d". That upgrade is false, and it is worth seeing why, because the failure is already visible in **Set**.

The category of $\triangleleft$-comonoids is, by [3, Cor. 5.12, Def. 5.11], isomorphic to the category of internal categories and internal *cofunctors* — written $\mathbf{Cat}^\sharp$ over **Set**. A cofunctor $F : C \to D$ goes forward on objects and *backward* on morphisms, lifting each D-arrow out of $F(x)$ to a C-arrow out of x. In $\mathbf{Cat}^\sharp$ the product category $C \times D$ is *not* the categorical product; it is a monoidal, non-cartesian product, coinciding with the parallel product [2, §8.2.4]. Theorem 1.1 is the internal, general-$\mathcal{E}$ form of this monoidal product, not of a categorical product.

Example 4.1 (The cone exists; universality fails)**.** Work in **Set**. Let $C = D = \mathbf{2} = \{0 \to 1\}$, the walking arrow, and let E be the category with objects e, x, z and non-identity arrows $u : e \to x$, $v : e \to z$. Define cofunctors $F, G : E \to \mathbf{2}$ by $e \mapsto 0$, $x, z \mapsto 1$, with F lifting the arrow $a : 0 \to 1$ at e to u and G lifting it to v. (The cofunctor laws hold trivially: there are no composable non-identity pairs.) The projections from the monoidal product, $\pi_C = \mathrm{id}_c \otimes \varepsilon_d$ and $\pi_D = \varepsilon_c \otimes \mathrm{id}_d$, are the cofunctors $(f, h) \mapsto f$ and $(f, h) \mapsto h$ on morphisms; so (F, G) is a cone. But any cofunctor $H : E \to \mathbf{2} \times \mathbf{2}$ with $\pi_C H = F$ and $\pi_D H = G$ must have $H(e) = (0, 0)$ and $H(x) = H(z) = (1, 1)$, and must then lift $(a, \mathrm{id}) : (0, 0) \to (1, 0)$ to an arrow out of e whose codomain maps to $(1, 0)$. No such arrow of E exists. So $(c \otimes d, \pi_C, \pi_D)$ is not a categorical product in $\mathbf{Cat}^\sharp$.

The genuine categorical product of C and D lives on the *functor* side. Internal categories in $\mathcal{E}$ are models of the finite-limit sketch $T_{\mathbf{Cat}}$, so for finitely complete $\mathcal{E}$ the category $\mathbf{Cat}(\mathcal{E}) = \mathrm{Mod}(T_{\mathbf{Cat}}, \mathcal{E})$ has all limits $\mathcal{E}$ has, computed objectwise; in particular the product $C \times D$ exists, with $(C \times D)_0 = C_0 \times D_0$, $(C \times D)_1 = C_1 \times D_1$, structure componentwise, and projections internal *functors*. (For a presheaf topos $\mathcal{E} = [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ this is $\mathbf{Cat}(\mathcal{E}) \simeq [\mathcal{C}^{\mathrm{op}}, \mathbf{Cat}]$ with the pointwise product.) This is the universal property one wants — but it is a fact about $\mathbf{Cat}(\mathcal{E})$, proved by the sketch argument, and it is *not* a statement about the tensor $\otimes$ on comonoids. Theorem 1.1 supplies the bridge: the *object* $c \otimes d$ carries exactly the product internal category, even though the *morphism* theory of comonoids sees $\otimes$ as a non-cartesian monoidal product.

5. What transports, and what the base must supply

The proof of Theorem 1.1 uses the base only through the ingredients in Table 1. The last column records the *weakest* property that suffices for that line.

ingredient	base property used	holds in
$\mathrm{Poly}(\mathcal{E})$, $\triangleleft$, $\otimes$, duoidality	exponentiable direction bundles [3]	any finitely complete $\mathcal{E}$ (exponentiable maps)
underlying object $c \otimes d = (C \times D)^\sharp$	finite-limit products	any finitely complete $\mathcal{E}$
interchanger coherence ($\star$)	products commute with pullbacks	any finitely complete $\mathcal{E}$
counit $\varepsilon = \mathrm{id} : C_0 \to C_1$	*none* (an internal section)	any $\mathcal{E}$ (no well-pointedness)

Table 1. The base appears only here. Beyond the Shapiro–Spivak framework (row 1), nothing exceeds "finitely complete".

Three properties one might have expected are *not* used: local cartesian closure (row 1 asks only for exponentiable direction bundles, which an LCCC supplies but does not require); extensivity or disjointness of coproducts (the coherence ($\star$) is a limit–limit interchange, not an extensivity axiom); and well-pointedness (the counit is the internal section id, Remark 2.3). In particular *no coproducts at all* are consumed by the identification.

6. A witness in **Graph**, as an illustration

As a finite sanity check we exhibit Theorem 1.1 in the presheaf topos $\mathcal{E} = \mathbf{Graph} = [\bullet \rightrightarrows \bullet, \mathbf{Set}]$ of directed graphs. Two honest caveats, following the referee: in a presheaf topos the internal-category side is computed *pointwise*, so this check cannot probe the genuinely non-pointwise content of $\triangleleft$ (that is discharged by the general proof, not here); and the "no global element" point needs no test at all, since the counit's datum is the morphism id and global elements never enter. The witness is therefore an illustration of the finished theorem in a concrete topos, not evidence for it.

By $\mathbf{Cat}(\mathbf{Graph}) \simeq [\bullet \rightrightarrows \bullet, \mathbf{Cat}]$, an internal category C in **Graph** is a diagram of two ordinary categories $\mathbf{C}_V, \mathbf{C}_E$ with two functors $S, T : \mathbf{C}_E \to \mathbf{C}_V$; its object-of-objects C_0 is the graph with vertex set $\mathrm{ob}\mathbf{C}_V$, edge set $\mathrm{ob}\mathbf{C}_E$ and source/target S_0, T_0, and the internal structure maps $\mathrm{dom}, \mathrm{cod}, \mathrm{id}, \mathrm{comp}$ are graph morphisms — commuting with the graph source/target s, t *iff* S, T are functors.

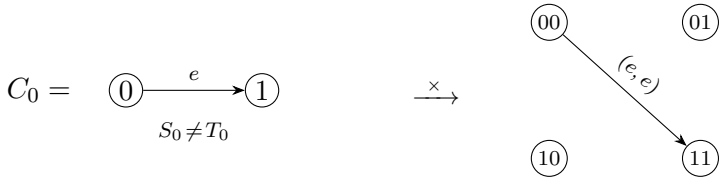


Figure 1. The illustrative witness `Carr`: the object-of-objects graph $C_0 = (0 \xrightarrow{e} 1)$ is loopless ($S_0 \neq T_0$), hence has no global element $1 \to C_0$. The product graph $(C \times C)_0 = C_0 \times C_0$ is computed *pointwise* in **Graph**: four vertices and a single edge $(e, e) : 00 \to 11$, again loopless, so again no global element. The identity-section and composition of $c \otimes c$ are nonetheless genuine graph morphisms.

We machine-check, in Lean 4 (core only, Mathlib-free), that for the three internal categories `Cpt` (the point, a loop — the baseline positive control, *with* a global element), `Carr` ($C_0 = (0 \to 1)$, loopless) and `Cdisc` (C_0 two loopless vertices), and all nine ordered pairs, the Dirichlet tensor $c \otimes d$ computed fibrewise and glued by $S \times S, T \times T$ is a genuine internal category in **Graph**: both fibres satisfy the category (= comonoid) axioms, and the glued source/target are functors — i.e. the identity-section and composition of $c \otimes d$ commute with s, t. The headline certificate is

`all_prod_valid : allProdValid = true` (by `decide`),

with the loopless case isolated as `prod_st_functors_Carr_Carr` together with `prod_loopless_Carr_Carr`. The negative controls `bogus` (a non-functor) and `notcat` (a non-category) are rejected by the same predicates. Every certificate reports [`propext`] only — no `sorryAx`, no `native_decide`. The file is https://github.com/scotmacbeth/work-in-progress/blob/main/lean/2026-10-21-day-convolution-grap lean; the **Set**-side base case is checked independently by a Python verifier (16/16 ordered pairs over four test categories, on the nose).

7. Discussion: honest bounds

The maximal reach is the Shapiro–Spivak setting. The identification uses, beyond finite limits, only that the polynomial framework of [3] is available — i.e. that the direction bundles are exponentiable. That is strictly weaker than local cartesian closure, and needs no coproducts: the dependent sums Σ_f are postcomposition, which exists in any category. So there is no "open upper bound" to flag: the honest maximal reach is the Shapiro–Spivak setting itself — any finitely complete category with exponentiable direction bundles — which includes every locally cartesian closed category and every elementary topos, Grothendieck or not, with or without a natural-numbers object.

"Day convolution" is Set-level motivation only. Over **Set**, the tensor $\otimes$ is genuinely a Day convolution: [2, Prop. 3.79] is a co-Yoneda coend computation in **Set**. Over a general $\mathcal{E}$ we make *no* Day-convolution claim. Stating one would require saying Day convolution *of what* (internal functors on $\mathcal{E}$? $\mathcal{E}$-enriched?), and the construction would need coends over a large category with copowers by internal homs and pullback-stability — genuinely base-sensitive data, and exactly where an elementary topos (no infinite coproducts) might differ from a Grothendieck one. Our proof never touches any of this: $\otimes$ is used only as the product of bundles [3, Def. 4.1]. This is why the title says "Dirichlet tensor".

No converse packaged in. We do *not* assert any converse of the form "these hypotheses force $\mathcal{E}$ to be a topos". The theorem is stated directly over finitely complete categories with exponentiable direction bundles, and §5 records which strictly weaker properties each line consumes.

What this buys the chain. The equivalence containers $\simeq \cdots \simeq$ **Cat** now carries its object-level monoidal content — the Dirichlet tensor represents the product internal category — uniformly across every finitely complete base the framework admits. A compositional model built in presheaves on a site, or sheaves on a space, inherits the product structure of its directed shapes with no recomputation and no appeal to global elements. The one thing it does *not* inherit for free is a universal property *among comonoids*: there the tensor is a monoidal, non-cartesian product (§4), and the categorical product lives on the functor side $\mathbf{Cat}(\mathcal{E})$. Keeping those two apart is the main honest correction this note makes to the folklore.

References

[1] N. Gambino and J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 1, 153–192; arXiv:0906.4931.
[2] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, book draft, 2023; arXiv:2312.00990.
[3] B. Shapiro and D. I. Spivak, *Structures on Categories of Polynomials*, 2023; arXiv:2305.00167.
[4] D. I. Spivak, R. Garner, and A. Fairbanks(-Lyons), *Functorial Aggregation*, 2021; arXiv:2111.10968.

Kodamai